Zhao et al. BMC Genomics (2025) 26:637
https://doi.org/10.1186/s12864-025-11834-0

RESEARCH

Open Access

Genome-wide identification and expression analysis of the *HvGATA* gene family under abiotic stresses in barley (*Hordeum vulgare* L.)

Yan Zhao[1], Chengxin Yin[1], Ruixi Xing[1], Jianbin Zeng[1], Xuehuan Dai[1], Dengan Xu[1], Wenxing Liu[1*] and Wujun Ma[1,2]

Abstract

Background GATA transcription factors play crucial roles in plant growth and development, especially in response to environmental stress. Although GATA genes have been studied and identified in various plants, research on these genes in barley is relatively limited.

Results This study identified the *GATA* gene family and analyzed its gene structure, chromosome distribution, evolutionary analysis, and expression patterns of the *HvGATA*gene family in barley. The results showed that 27 *HvGATA* genes are unevenly distributed across seven chromosomes and divided into four subfamilies with similar structures within the same subfamily. Gene synthesis analysis revealed that *HvGATA* gene family has undergone significant purifying selection. It is noteworthy that the promoter regions of *HvGATA* genes displayed many cis-acting elements associated with stress responses and hormone regulation. Additionally, the 27 identified genes are predominantly involved in responses to inorganic substances, as indicated by the Gene Ontology (GO) enrichment analysis. The majority of miRNAs that regulate these genes are also capable of modulating abiotic stress responses. Furthermore, expression analysis confirms that the majority of *HvGATA* genes participate in the regulation of abiotic stresses.

Conclusion In summary, this study contribute to our understanding of important role of *HvGATAs* in barley, providing a foundation for further exploration of gene function and target genes related to stress responses.

Keywords Genome-wide, Barley (*Hordeum vulgare* L.), GATA, Expression patterns, Regulatory network

Introduction

During their growth and development, plants are exposed to various environmental stresses, especially abiotic stress conditions. Therefore, throughout the course of evolution, many plants have developed various protective mechanisms to adapt to these stresses. Among these mechanisms, transcriptional regulation is essential in governing the process. Transcription factors (TFs) are specialized proteins that regulate the expression of downstream genes by specifically binding to their promoter regions, influencing gene expression and controlling various critical biological processes including signal transduction, cell morphogenesis, and responses to environmental stresses [1, 2]. To date, numerous transcription factor families have been identified in plant research, such as *GATA*, *MYB*, *DREB*, *bZIP*, *WRKY and MADS-box* [3].

GATAs are a family of DNA-binding proteins that are widely present in fungi, animals, and plants. They belong to the zinc finger family of type IV, their DNA-binding

*Correspondence:
Wenxing Liu
liuwx@qau.edu.cn
[1]College of Agronomy, Qingdao Agricultural University, Qingdao 266109, China
[2]School of Agriculture, Murdoch University, Perth, WA 4350, Australia

domain contains a $CX_2CX_{17-20}C$-X_2C motif. NTL1 discovered in tobacco (*Nicotiana tabacum*) is the first GATA factor, and it is a homolog of NIT2 from *Neurospora crassa* [3].

The *GATA* family genes have been discovered across various plant species, including Arabidopsis, rice, purple falsebrome, pepper, cucumber, and cotton [3]. With the rapid advancement of next-generation sequencing technologies, *GATA* gene families have been discovered in both monocots and dicots. According to genome-wide analyses, Arabidopsis contains 30 *GATA* genes [4] rice has 28 [4], apple possesses 35 [5], and soybean has 64 *GATA* genes [6], These genes are involved in various functions, including the regulation of seed dormancy, host immune responses, photosynthesis and growth, chloroplast development, grain shape, and responses to abiotic stresses [3]. For example, *AtGATA12* is negatively regulated by the DELLA protein RGL2, thereby affecting seed dormancy [7]. In rice, over-expression of *OsGATA12* led to reduced leaves and tillers, thereby influencing yield-associated traits [8]. *OsGATA7* enhances brassinosteroid signaling and partially mediates brassinosteroid-regulated plant growth in certain processes, thus achieving the objective of brassinosteroid-mediated regulation of plant architecture and seed morphology [9]. In soybean, nitrogen deficient treatment significantly reduced the expression of *GATA44* and *GATA58* in seedlings [6]. In poplar, *PdGATA19* plays a key role in photosynthesis and growth [8]. In sweet potato, IbGATA24 interacts with the COP9-5 A protein to regulate salt and drought tolerance [10]. In wheat, over-expression of *TaZIM-A1* can delay flowering time, leading to reduced thousand-grain weight [11]. Studies have also shown that over-expression of *TaGATA1* can enhance resistance to *Fusarium graminearum* in wheat [12]. When plants are subjected to abiotic stress, the promoters of genes will interact with transcription factors, which in turn upregulates the gene expression levels. Through a series of intricate physiological and molecular processes, the plants' stress tolerance is enhanced. For example, it can regulate the ATP synthesis process, affect the sugar metabolism process [13], or regulate osmoregulatory proteins [14], etc., to enhance the plant's adaptability to adverse conditions. In summary, GATA factors are powerful and important transcription factors involved in regulating many vital physiological and biochemical processes during plant growth and development.

Barley is the fourth largest cereal crop globally with a long history of cultivation [15]. On the one hand, barley can be used as a staple food, for brewing and medicinal purposes, making it highly valuable. On the other hand, as a traditional crop, barley has a wide distribution and long cultivation history in China, which has broad ecological adaptability. The early research on plants was generally focused on model plants, such as *Arabidopsis thaliana* and *Nicotiana tabacum*. In recent years, with the continuous development of whole-genome sequencing technology and the enhancement of sequencing depth, which provides us the possibility to understand the evolutionary development of gene families at the genomic level. Furthermore, barley is closely related to wheat. Compared with model crops such as *Arabidopsis thaliana* and *Brachypodium distachyon*, barley is a more suitable model crop for conducting research on the gene functions of wheat, which further highlights the significance of the research on barley.

This study performed an extensive genome-wide analysis of *GATA* genes in barley. 27 candidate *HvGATA* genes through bioinformatic analysis of the barley genome were identified. It characterized their phylogenetic tree, gene structure, protein motifs, chromosome distribution, cis-element analysis, interaction with miRNAs, prediction of regulated genes, and *GATA* expression in different tissues, abiotic stressed, which can provide important guiding role for the subsequent functional research of *GATA* genes in barley.

Materials and methods

Identification of *HvGATA* genes in barley

The gene and protein sequences of barley were obtained from the Ensemble Plants (accessible at https://plants.ensembl.org/index.html) [16]. The barley genome prot ein database was screened using HMMER 3.0 [17] with a Hidden Markov Model (HMM) search, employing the GATA protein domain PF00320 as the default parameter. The sequence of the longest transcript within each gene ID was retained, and sequences without start or stop codons were discarded. The remaining sequences were analyzed using Pfam tools with an e-value threshold of < e-20 and further examined using the Conserved Domain Database (CDD) [18]. Ultimately, 27 *HvGATA* genes were identified. Amino acid count, isoelectric point (pI), and the molecular weight (MW) were determined using the ExPASy tool (https://www.expasy.org/) [19].

Phylogenetic analysis of *HvGATAs*

The sequences of 27 HvGATA proteins were aligned using the ClustalW program [20]. The Neighbor-Joining method was applied to construct a phylogenetic tree using MEGA 7.0 software [21], with the parameters configured to the Poisson model, pairwise deletion, and 1000 bootstrap replicates. The phylogenetic tree was constructed and visualized using the online tool iTOL (https://itol.embl.de/) for enhanced presentation [22].

Chromosome localization and gene duplication

The MapChart tool (v2.3.2) was utilized for visualizing the chromosome localization of *HvGATA* genes [23].

MCScanX software was utilized to examine the collinearity of orthologous *GATA* genes between barley, Arabidopsis, maize, rice, and wheat [24]. The ParaAT tool was employed to compute the the rate of non-synonymous substitutions (Ka), the rate of synonymous substitutions (Ks), and the ratio of Ka: Ks [25]. The Calculator 2.0 tool was utilized to perform the calculations [26]. Additionally, The gene duplication time was estimated using the formula $T = Ks/2\lambda$, $\lambda = 1.5 \times 10^{-8}$ [26].

Gene structure and protein motif analysis

The GSDS (Gene Structure Display Server) tool (http://gsds.cbi.pku.edu.cn/) was applied to analyze the exon-intron structure of *HvGATA* genes [27]. Conserved motifs in HvGATA proteins were identified using the MEME suite (https://www.ebi.ac.uk/jdispatcher/) [28]. Exon-intron organization and conserved motif analysis of HvGATA were conducted using the GFF3 database and TBtools from Ensemble Plants [29].

Cis-element analysis in *HvGATA* promoter regions

Promoter sequences (−1500 bp) of *HvGATA* genes were retrieved from the barley genome sequence, and the cis-elements in the promoter regions were analyzed using PlantCARE database (http://bioinformatics.psb.ugent.be/webtools/plantcare/html/) [30].The TBtools was used to annotate the complete graphics of cis-elements [29].

Prediction of regulated genes by *HvGATA*

To gain deeper insights into the regulatory mechanisms of *HvGATAs*, the online tool PlantRegMap was used to predict target genes regulated by *HvGATAs* [31], with species settings for barley, Organ settings for All, Method settings for FunTFBS, Mode settings for TF (search for binding sites). The Gene Ontology (GO) enrichment analysis was conducted using an online bioinformatics server (http://www.bioinformatics.com.cn/).

Prediction of miRNA-*HvGATA* regulatory networks

The psRNATarget server was used to predict miRNAs capable of regulating *HvGATA* gene expression, employing the following parameters: G:U pair penalty score of 1, allowing zero mismatches in the seed region [32]. The interaction network was displayed using Cytoscape V3.8.2 software (https://cytoscape.org/download.html) [33].

Expression profiling analysis of *HvGATAs*

The expression patterns of *GATA* genes in different barley tissues were obtained from the WheatOmics website (http://wheatomics.sdau.edu.cn/) [34]. Gene expression values were represented as Transcripts Per Million (TPM) per kilobase of exon per million reads. The average expression levels, derived from three biological replicates, were used to depict the expression patterns across various tissues. The data were normalized using the expression level in roots as a reference. The quantitative real-time PCR (qRT-PCR) experiment was conducted according to Livak and Schmittgen (2001) [35] with wild barley XZ5 as the material and subjected to drought and salt stress treatments with 25% PEG2000 and 20% NaCl, respectively. Sampling roots and leaves at 0 h, 6 h, and 24 h treatment. The primers were given in Table S16.

Results

Identification and distribution of the *HvGATA* gene family in barley

A total of 27 *HvGATA* genes were identified in this study and named according to their locations on the chromosomes as *HvGATA1* to *HvGATA27* (Fig. 1). Chromosome 4 contains the most number of *GATA* genes (five), followed by chromosomes 1 and 2 with four each, chromosomes 3 and 6 with three each, and chromosomes 5 and 7 with two each. The gene names, IDs, chromosome locations, number of amino acids, molecular weights, and isoelectric points of these genes were organized (Table S1). The protein sequence lengths of HvGATA ranged from 155 to 775 amino acids, with molecular weights ranging from 17.34 to 88.61 KDa (Table S2).

Phylogenetic analysis of HvGATA proteins

To explore the phylogenetic relationships of *HvGATAs*, a phylogenetic tree was constructed based on 27 barley *HvGATA* genes and 30 Arabidopsis *AtGATA* genes (Fig. 2). The AtGATA protein sequences were listed in Table S3. According to previous reports, the 30 AtGATA proteins can be divided into four subgroups, thus barley *HvGATAs* can also be classified into four subgroups. Subgroup I included 11 *HvGATA* and 14 *AtGATA*; Subgroup II included 7 *HvGATA* and 11 *AtGATA*; Subgroup III contains 5 *HvGATA* and 3 *AtGATA*; Subgroup IV included 4 *HvGATA* and 2 *AtGATA*.

HvGATA structural analysis

To further understand the gene structure characteristics of barley *HvGATA*, its DNA structure was analyzed (Fig. 3A). Among the 27 *HvGATA* genes, the number of introns is unevenly distributed, ranging from 0 to 7. Genes within the same subfamily generally have similar numbers of introns. Specifically, *HvGATA2*, *HvGATA18*, *HvGATA4*, *HvGATA11*, *HvGATA25*, *HvGATA7*, and *HvGATA24* in subfamily I, as well as members of subfamilies II and III, excluding *HvGATA1*, contain 0 to 2 introns. In contrast, other *HvGATA* genes in subfamily I and those in subfamily IV contain 4 to 7 introns. Moreover, 10 motifs were identified within *HvGATA* and named motif1-10 (Table S4). All *HvGATA* genes contain

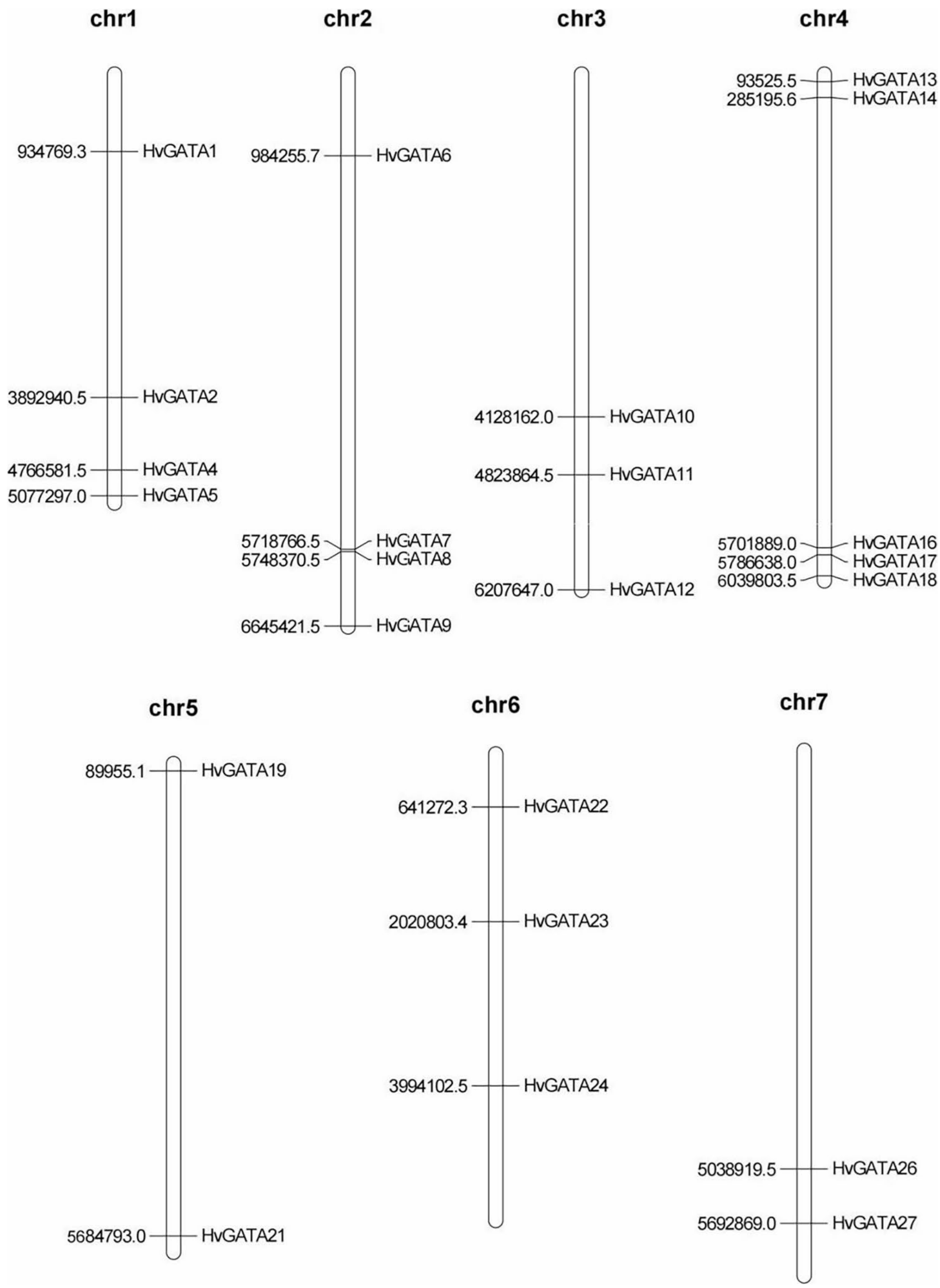

Fig. 1 Distribution of *HvGATA* genes in barley chromosomes. The chromosome numbers are indicated at the top of each chromosome image

a conserved zinc finger domain motif 1. Those with only one motif 1 are classified into subfamily I. In addition to *HvGATA1*, genes containing motif 3 and motif 5 are also placed into subfamily I. Genes containing both motif 2 and motif 6 are classified into subfamily II and subfamily III. Although subfamily II and subfamily III share the same motifs, all members of subfamily III contain only one motif 1, one motif 2, and one motif 6. In contrast, most members of subfamily II contain duplicated motif segments. Genes containing motifs 7, 8, 9, and 10 are classified into subfamily IV (Fig. 3B).

Analysis of cis-elements in the promoters of *HvGATAs*

To gain insight into the functions and regulatory patterns of *HvGATA* genes, the promoter sequences of 27 *HvGATA* were scanned and analyzed for their cis-elements such as ABRE, AuxRR-core, ARE, DRE, GT1-motif, MBS, SARE, Sp1, TCA, TGA, and TC-rich repeats. These cis-elements are associated with processes like ABA response, auxin regulation, drought tolerance, salicylic acid response, stress resistance, and cold stress response (Fig. 4; Table S5). In summary, among the 27 *HvGATA* genes, 25 contain the ABRE element, 3 contain

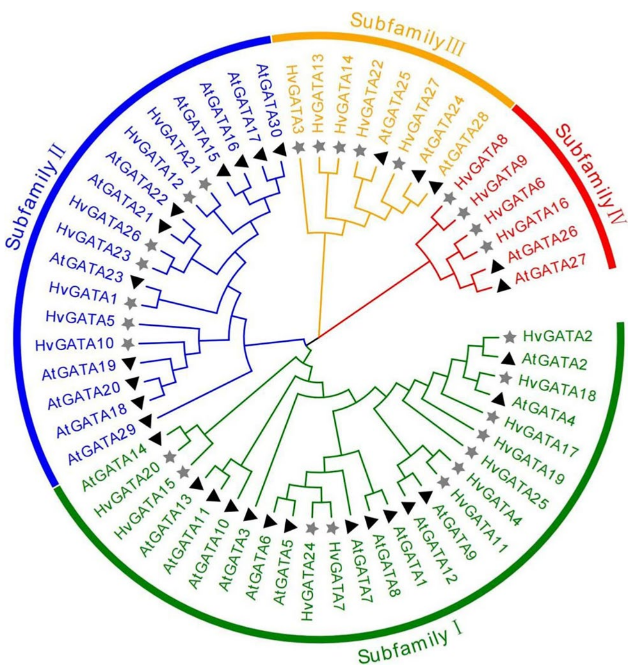

Fig. 2 Phylogenetic tree of full-length HvGATA and AtGATA proteins. Using the Maximum Likelihood method based on the JTT matrix-based model. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site. All positions containing gaps and missing data were eliminated. The different colored arcs indicate subfamilies of the GATA proteins. Different colored shapes represent GATAs from barley (☆) and Arabidopsis (▲)

AuxRR-core, 17 contain ARE, 1 contains DRE, 7 contain GT1-motif, 13 contain MBS, 1 contains SARE, 13 contain Sp1, 9 contain TCA, 8 contain TGA, and 6 contain TC-rich repeats.

Tissue-specific expression patterns of *HvGATAs*

To further understand the expression of *HvGATA* in barley plants, the transcription levels of *HvGATA* in different tissues at various developmental stages were compared. Based on the expression differences, the 27 *HvGATA* were divided into three groups (Fig. 5; Table S6). The first group included only *HvGATA8* and *HvGATA9*, the second group included *HvGATA6* and *HvGATA16*, and the remaining 23 *HvGATA* can be further subdivided into two subgroups: *HvGATA27*, *HvGATA22*, *HvGATA14*, *HvGATA13*, and *HvGATA3* make up subgroup one, while the rest form subgroup two. Even within the same subfamily, gene expression levels vary.

Expression patterns of *HvGATAs* under different stresses

Based on transcriptomic data from published papers, the expression levels of *HvGATA* genes under abiotic stresses such as low phosphorus, drought, high temperature, low temperature, and salt stress were analyzed (Fig. 6; Table S7). These results revealed significant differences in the expression levels of *HvGATA* genes under various stress conditions. *HvGATA*4/7/11/14/20/21/25 exhibited higher expression levels under low-temperature stress, while genes such as *HvGATA*2/18/19 showed reduced expression under the same condition. Additionally, most *HvGATA* genes were involved in the high-temperature stress response, with their expression levels decreasing. Under low phosphorus stress, the expression of *HvGATA9* and *HvGATA12* increased significantly, whereas the expression of *HvGATA2* and *HvGATA26* decreased.

The data given above were consistent with the qRT-PCR experiment. e.g. The expression levels of *HvGATA9*, *HvGATA16*, and *HvGATA20* increased 26%, 80% and 122% under drought stress, while the expression level of *HvGATA1* decreased 41% (Fig. 7). The expression levels of *HvGATA20* and *HvGATA24* increased 207% and 316% under salt stress, whereas the expression levels of *HvGATA13* and *HvGATA14* decreased 52% and 39% (Fig. 7). This experiment enhanced the reliability of the expression analysis results.

Analysis of *HvGATAs* targeted genes and miRNA targeting *HvGATAs*

GO enrichment analysis was performed on downstream genes targeted by barley *HvGATAs*, revealing that six target genes are involved in inorganic substance responses, four in extracellular region regulation, and four in unfolded protein binding (Fig. 8; Table S9). To explore the potential miRNA-mRNA regulatory networks in barley, a scan was conducted on the known 321 miRNAs. Finally, it was predicted that 19 miRNAs could inhibit the transcription levels of *HvGATA* (Fig. 9; Table S10). In summary, these 19 miRNAs were classified into 11 groups: Group 1 included four miRNAs, targeting *HvGATA15* and *HvGATA16*; Group 2 included three miRNAs, targeting *HvGATA6*; Group 3 included two miRNAs, targeting *HvGATA18*; and Groups 4 to 11 involved in *HvGATA5*, *HvGATA8*, *HvGATA12*, *HvGATA14*, *HvGATA19*, *HvGATA24*, *HvGATA25*, and *HvGATA27* respectively (Fig. 10).

Discussion

Transcription factors participate in the regulation of various physiological processes throughout plant ontogeny and development. To date, the *GATA* gene family has been identified in a wide range of plant species, with varying numbers of *GATA* genes across species. For instance, Arabidopsis has 30 *GATA* family members, wheat has 79, apple has 35, rice contains 28, and Brassica napus contains 96 [3]. This variation in gene numbers may be attributed to differences in genome size and complexity across species. While substantial data on the gene structure, expression profiles, characteristics, and functions of *GATA* genes exist, a thorough

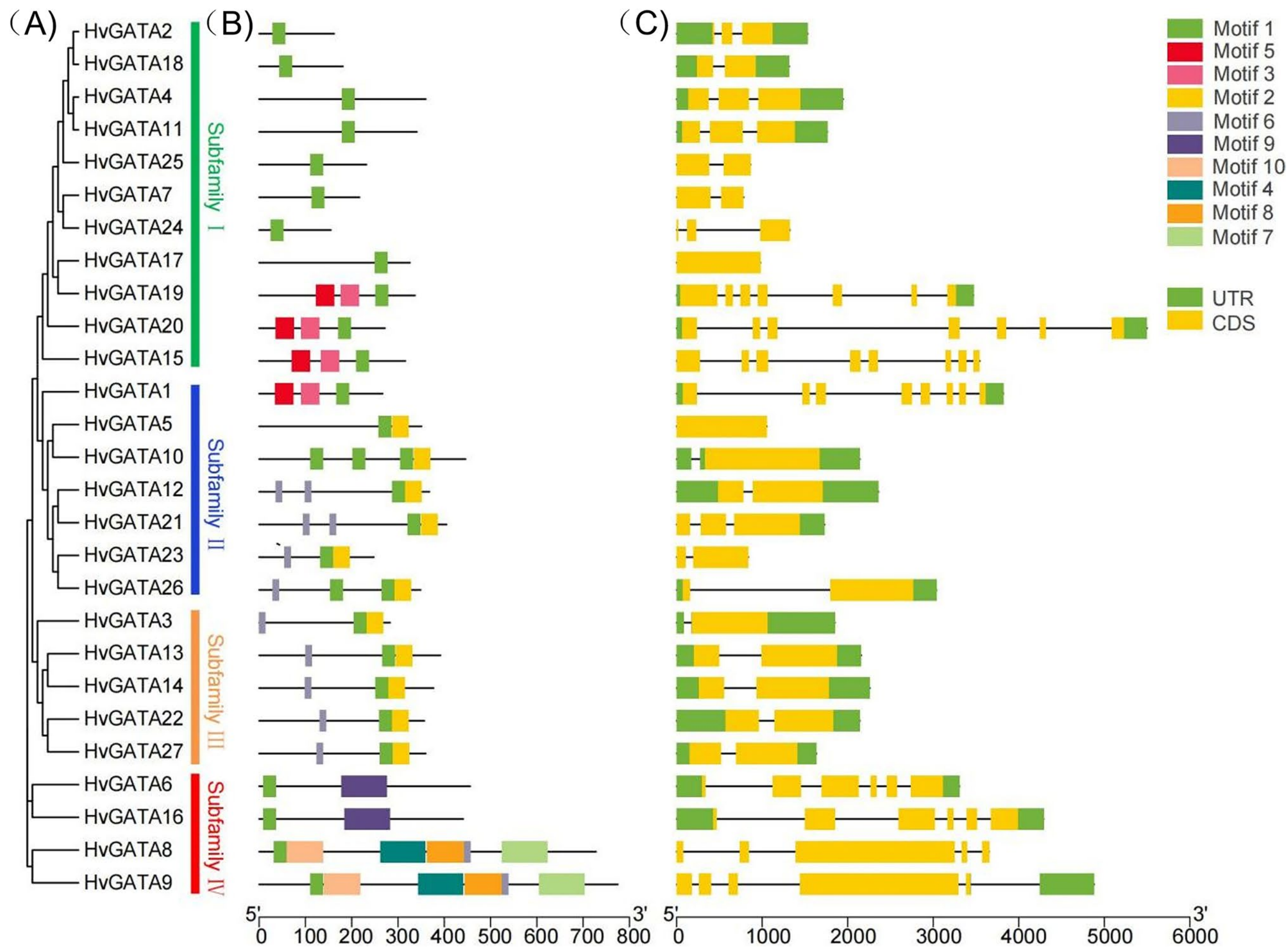

Fig. 3 Phylogenetic relationships, architecture of conserved protein motifs and gene structure in GATA genes from barley. **A** The phylogenetic tree was constructed based on the full-length sequences of maize GATA proteins using MEGA 7 software. **B** The motif compositions of 27 HvGATA proteins. The motifs were identified using the MEME program. Boxes of different colors represent motifs 1 to 10. The length of the amino acid sequences can be estimated by the scale at the bottom. **C** Gene structures of 27 *HvGATA* genes. Yellow boxes represent exons, green boxes represent 5′ or 3′ untranslated regions (UTR), and black lines represent introns. The length of nucleotide sequences of exons/introns/UTRs can be estimated by the scale at the bottom

genomic analysis of the *GATA* family in barley remains lacking. This research identified 27 members of the *HvGATA* gene family. within the barley genome, designated *HvGATA1* to *HvGATA27* (Fig. 1; Table S1). The 27 *HvGATA* genes were classified into four subfamilies, which aligns with grouping patterns found in other species, indicating the evolutionary stability of *GATA* family members in plants [6]. However, significant differences in genetic structure and expression patterns were observed among the subfamilies (Figs. 2 and 3). Subfamily IV contained unique motifs not present in other subfamilies, suggesting specialized functions. Furthermore, the number of introns varied among subfamilies, with subfamilies I and IV containing more introns, these genes may contribute to the evolution and functional diversification of *HvGATA* genes. Among these, 6 duplicated segments were identified, indicating that gene duplication has played a role in the expansion of the *GATA* family in barley, a synteny analysis between barley and other plants revealed significant collinearity, although not all *GATA* genes had one-to-one correspondence across species, further supporting the presence of extensive gene duplications within the *GATA* family. The genomic distribution of *HvGATA* genes exhibits distinct positional bias, with *HvGATA23* representing the sole exception through its mid-chromosomal localization, while all other family members consistently occupy subtelomeric regions. This non-random distribution pattern implies significant evolutionary constraints, particularly evidenced by the dense clustering of nine *HvGATA* genes within the terminal segments of chromosomes 1, 2, and 4. Such genomic architecture likely reflects strong selective pressures to preserve these regulatory elements for critical physiological functions and environmental adaptation mechanisms in barley. To comprehensively understand their evolutionary history, it is necessary to integrate evidence from other aspects for a comprehensive analysis. Ka/Ks analysis of gene duplications showed two groups

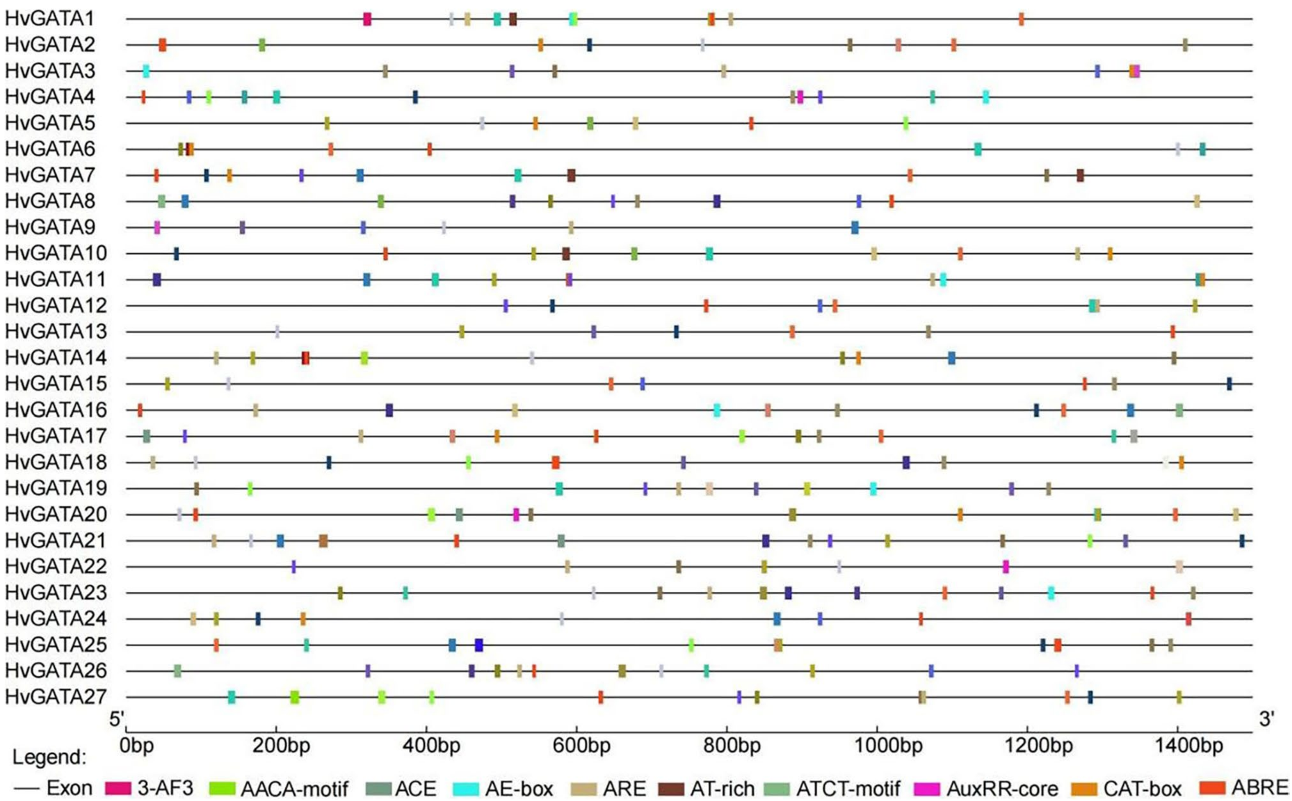

Fig. 4 Predicted cis-regulatory elements in *HvGATA* promoters. Promoter sequences (about 1.5 kb) of 27 *HvGATA* genes were analyzed by PlantCARE. The upstream length to the translation starting site can be inferred according to the scale at the bottom

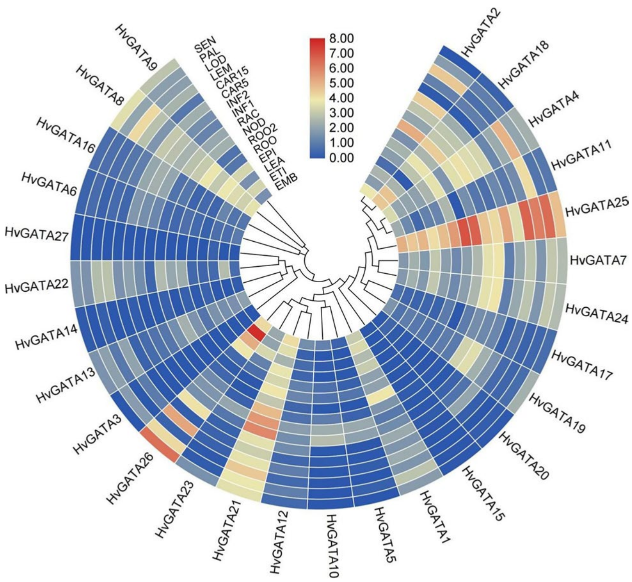

Fig. 5 Expression profiles of the *HvGATA* genes. Expression profiles of the *HvGATA* genes in different tissues. The color scale represents expression data with row scale. Blue: low expression; red: high expression

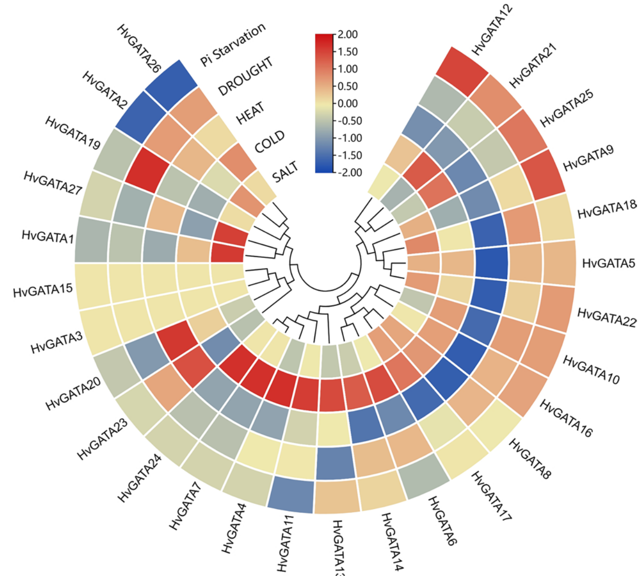

Fig. 6 Expression patterns of *HvGATAs* under different stresses. Expression data were the ratios to control values. The color scale represents expression levels from up-regulation (red) to down-regulation (blue)

with ratios greater than 1, indicating positive selection, while other groups exhibited ratios less than 1, suggesting purifying selection [36]. The collinearity analysis revealed that *GATA* genes in these species showing a one-to-one correspondence (Fig. 11). This suggests that *HvGATA* and *TaGATA* are highly conserved and that these two species share a close evolutionary relationship. The collinearity analysis between barley and maize *GATA* genes displayed

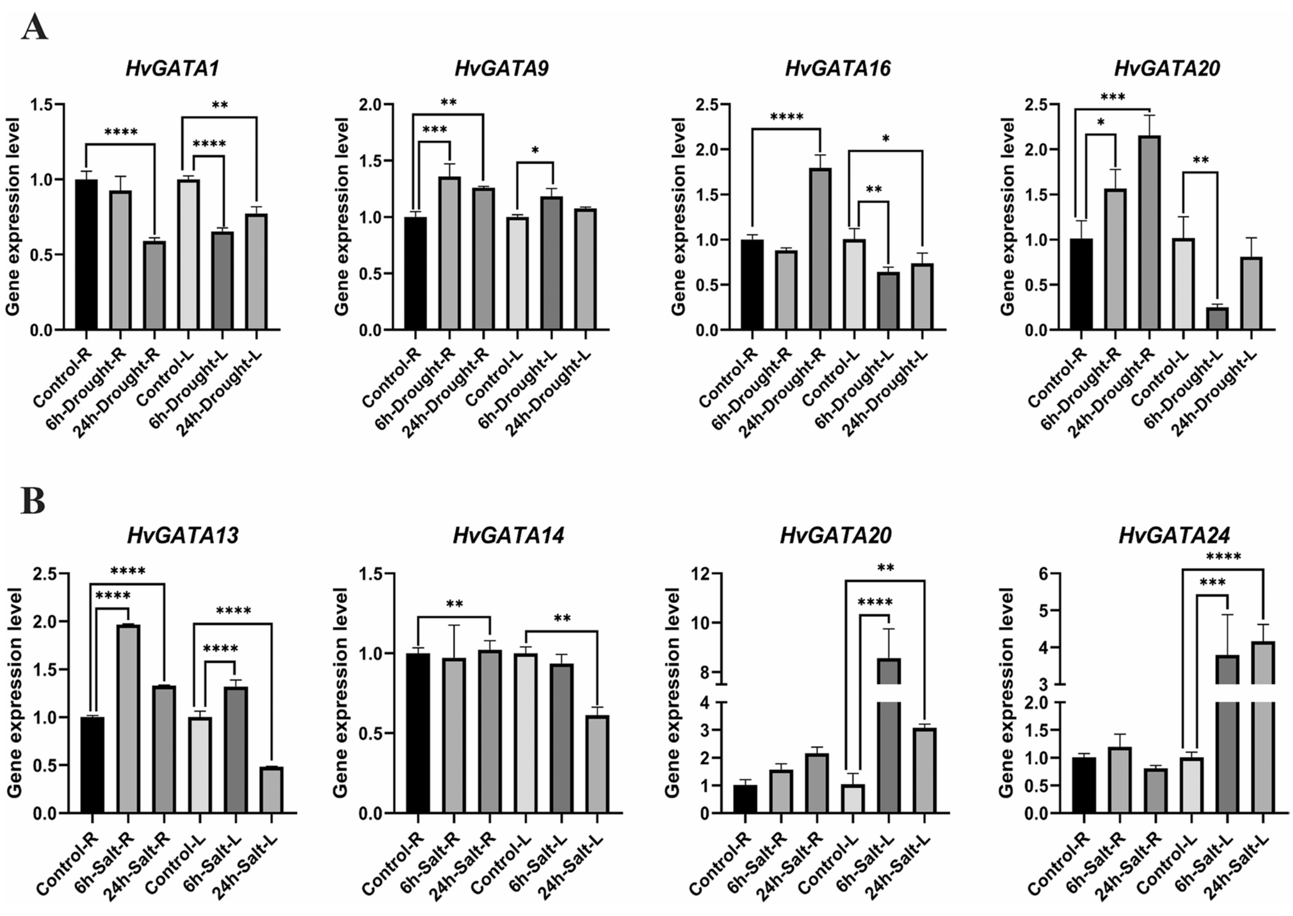

Fig. 7 Quantitative Real-Time PCR validation of gene expression under stress conditions. **A** Expression profiles of four genes after 6 h and 24 h of 25% PEG treatment. **B** Expression profiles of four genes after 6 h and 24 h of 20% NaCl treatment. R: Root; L: Leaf. The specific primers for PCR and the ACTIN sequence are listed in Supplementary Table S16

a more complex pattern, with most barley *GATA* genes having orthologous counterparts on different maize chromosomes. This suggests that maize *GATA* genes may have undergone duplication events during evolution, which is supported by evidence of whole-genome duplication in maize [37].

Cis-regulatory element analysis revealed that most *HvGATA* genes contained ABRE, ARE, and MBS elements, which are involved in regulating abiotic stress responses. The presence of these elements in the upstream regions of these genes may facilitate the binding of transcription factors, thereby regulating gene expression levels. This regulation can enhance stress resistance through various mechanisms, such as responding to the ABA signaling pathway, activating the antioxidant defense system, and participating in the synthesis of osmoregulatory substances [38, 39]. Additionally, three genes contain the AuxRR-core element. The upstream regions of genes with the AuxRR-core element may also bind transcription factors, thereby participating in auxin signal transduction and regulating plant growth and development processes [40]. Consistent with this, the overexpression of *TaGATA62* and *TaGATA73* in yeast and Arabidopsis resulted in improved tolerance to drought and salinity [41], and the *HvGATA22* and *HvGATA25* showed significantly higher expression under salt and drought stress. These findings suggest that *HvGATA* genes may regulate stress responses through the ABA signaling pathway. In phosphorus starvation conditions, *HvGATA26*, homologous to *TaGATA74*, *TaGATA76*, and *TaGATA78*, exhibited increased expression [3]. Furthermore, the GO enrichment analysis indicate that most genes are classified under "response to inorganic substances". When plants are exposed to abiotic stress, they often need to regulate the balance of intracellular inorganic ions to maintain normal physiological functions. This suggests that these genes may play a role in responding to abiotic stress [42]. Additionally, many genes are involved in the regulation of the extracellular region. Studies have shown that extracellular ATP is involved in signal transduction during stress responses. By binding to receptors on the cell surface, ATP activates downstream signaling pathways that help regulate plant stress resistance [43].

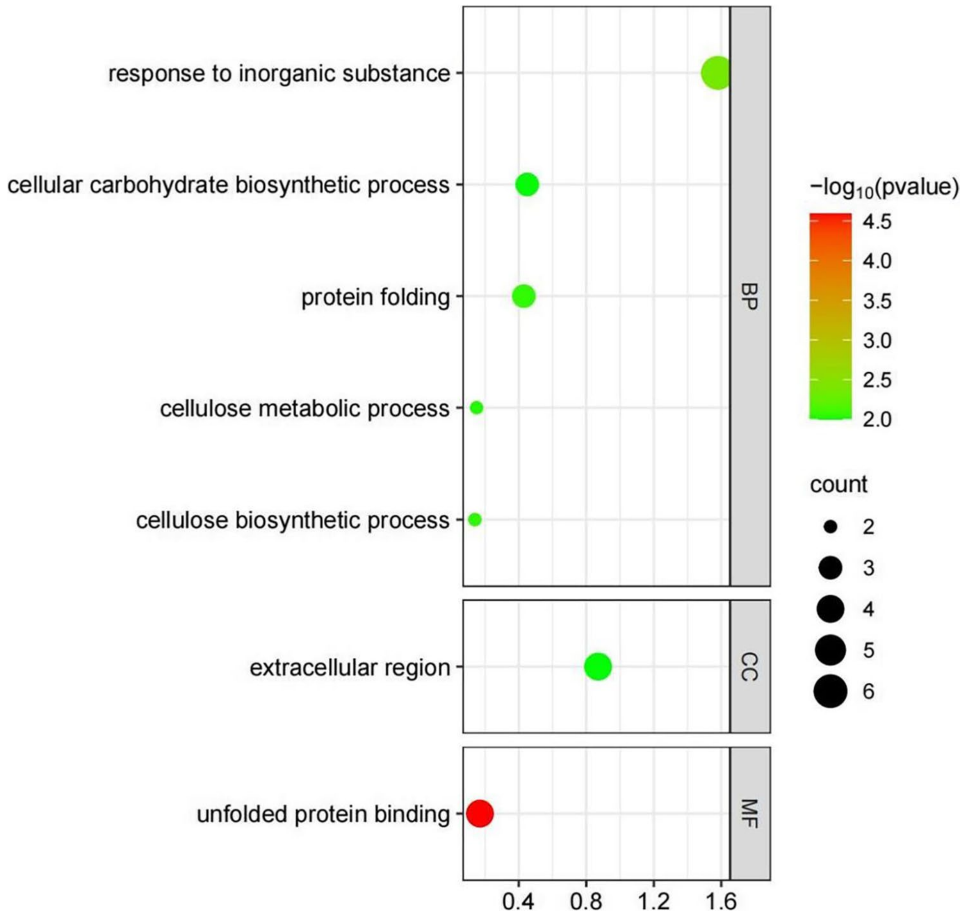

Fig. 8 Bubble map of GO enrichment of *HvGATA*-target genes. Genes were listed in Table S8. The X-axis represents Rich Ratio. Rich Ratio = Term Candidate Gene Number/Term Gene Number. The Y-axis represents GO Term. The size of the bubble represents the number of different genes annotated to a GO Term, and the color represents the enriched *P* value. $0 < P$ value < 1. The smaller the *P* value, the more significant the GO enrichment

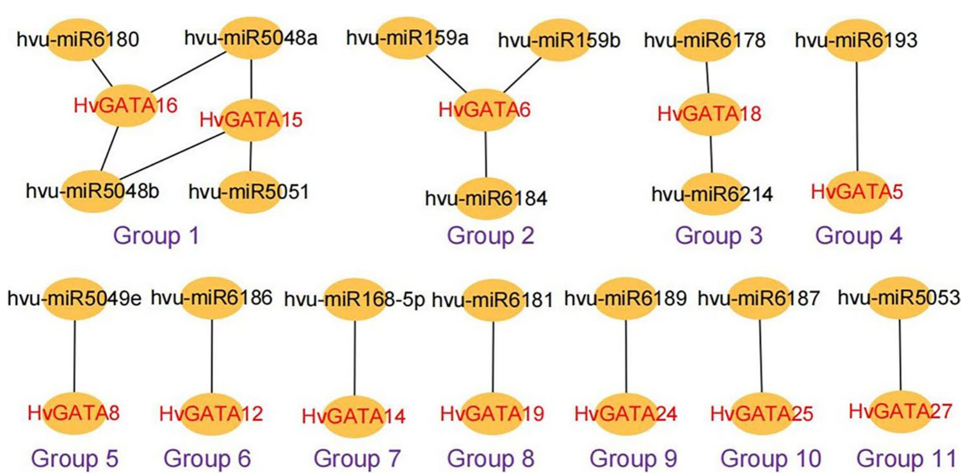

Fig. 9 Interaction networks between miRNAs and miRNAs-acted *HvGATAs*. Information on miRNAs and miRNAs-acted *HvGATAs* was shown in Table S10

The expression patterns of *HvGATA* genes, as analyzed through quantitative experiments, were consistent with the predicted results. This indicates that different *HvGATA* genes have specific expression patterns in response to various abiotic stresses. They may play distinct roles in barley's adaptation to diverse environmental stresses, helping the plant cope with complex and variable external environments through their respective expression regulation. However, under different stress conditions, the gene expression patterns varied across

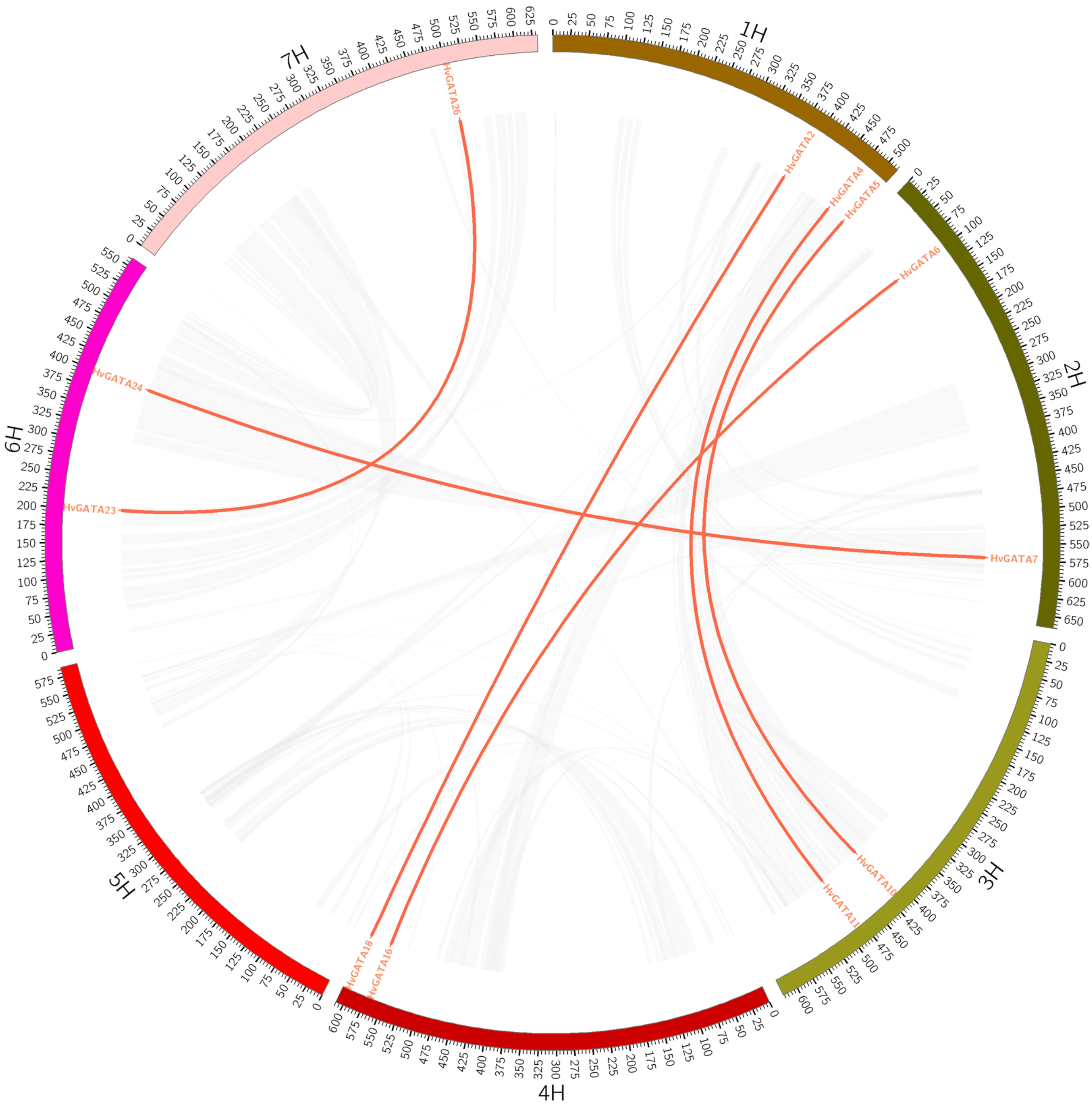

Fig. 10 Synteny analysis of *GATA* genes between barley, Arabidopsis, rice and wheat. Gray lines indicate all synteny blocks in the maize genome. The genes linked by red lines represent homologues

different tissues, with drought-tolerant genes predominantly expressed in the roots and salt-tolerant genes mainly expressed in the roots. This may be attributed to the fact that roots are the organs that first sense changes in water availability, and they need to promptly transmit signals to the aboveground parts. Additionally, roots need to undergo adaptive changes in morphology or structure to reduce water loss and enhance water use efficiency [44]. Leaves, being the primary organs for transpiration in plants, can also enhance their water storage capacity and reduce transpiration by altering their morphology or structure under salt stress, thereby mitigating the detrimental effects of salt stress [45]. These results further confirm the complexity and diversity of the *GATA* gene family in plant adaptation to environmental stresses. It is noteworthy that, although significant changes in gene expression were observed for some genes, the expression changes of other genes under stress treatment were not pronounced, and some genes exhibited specific expression discrepancies. For instance, under salt stress, the

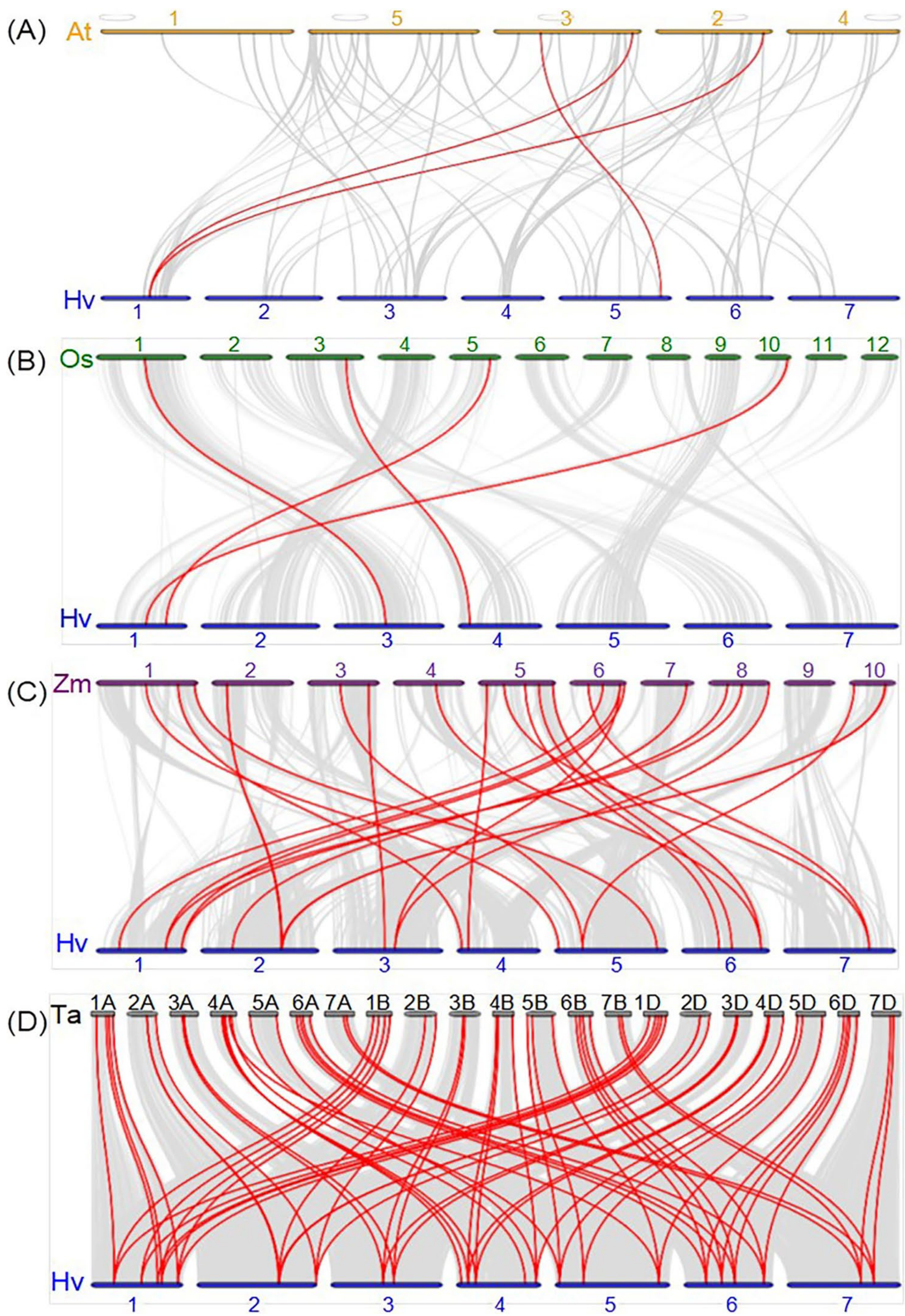

Fig. 11 Synteny analysis of *GATA* genes between barley, Arabidopsis, rice, maize and wheat. Gray lines: all collinear blocks within maize and other plant genomes. Red lines: the synteny of *GATA* gene pairs

expression level of *HvGATA13* increased in the roots but decreased in the leaves. This may be ascribed to the inherent characteristics of the experimental material and differences in its growth stages, or it may be related to the complex regulatory mechanisms of these genes under specific stress conditions.

The different subfamilies of *HvGATA* genes exhibit distinct functions and stress responses. Subfamily I genes typically have a higher number of introns and contain ABA-responsive elements (ABRE) and stress-responsive elements (MBS) in their promoter regions, suggesting their role in stress responses through the ABA signaling pathway, particularly in drought tolerance [38, 39].

Subfamily IV genes also have a relatively high number of introns and are likely involved in plant growth, development, and possibly photosynthesis [40]. Subfamilies II and III have fewer introns (usually 0–2) and their expression patterns under stress differ from those in subfamily I, indicating their involvement in other biological processes like growth regulation and hormone signaling. These differences reflect the functional diversification and evolutionary adaptation of the *HvGATA* gene family. Genes in the *GATA* subfamily I have been shown to be associated with plant growth and responses to environmental stresses [46]. In Arabidopsis, *BME3* (a homolog of *AtGATA25*) enhanced seed germination ability [46]. Compared to non-mutant plants, seed *BME3* gene knockout plants exhibited increased susceptibility to dormancy and cold stress, further validating the involvement of I subfamily genes in regulating stress resistance in barley. Studies found that the R-GL2-DOF6 complex regulates *GATA* subfamily I (*GATA12*), promoting dormancy of seed in Arabidopsis [47]. *GATA* genes of Subfamily II may be implicated in the regulation of flowering and are also involved in the plant's response to abiotic stresses [48]. In Arabidopsis, *GNC* and *GNL* (homologous genes of *HvGATA21* and *HvGATA26*) were related to germination, flowering, and the cold stress [49]. The expression level of *BnGATA2.5* in Brassica napus (a homolog of *HvGATA21* and *HvGATA26*) significantly decreases under ABA, drought, and cold stress [50]. In this study, *HvGATA26* was highly expressed in yellowing stem leaves and senescent leaves, with its promoter containing ABER, ARE, GT1-motif, and TGA cis-elements involved in abscisic acid, anaerobic induction, light response, and auxin regulation, it elucidates the correlation between abiotic stress and the *HvGATA26*. The promoter of *HvGATA21* harbored ABRE and G-box elements, which are associated with abscisic acid response, response to light, and drought responsiveness (Fig. 4;Table S5), suggesting a robust response to environmental stresses. Furthermore, overexpression of *OsGATA16* and *OsGATA8* in rice improved resistance to cold and drought stress, respectively [51, 52]. The overexpression of *SlGATA17* enhanced drought tolerance in tomato [53].

MicroRNAs (miRNAs) can target the 3' untranslated region (3'UTR) of their target mRNAs through complementary base pairing, leading to mRNA degradation or translational repression. When barley is subjected to abiotic stress, the corresponding miRNAs can target the 3'UTR of *HvGATA* genes in this manner, exerting negative regulation [54]. This mechanism can improve the stress resistance of the crop. Studies have identified hvu-miR6187 in the drought-tolerant rice variety Nagina22 [55], and in this study, we predicted that *HvGATA25* is regulated by hvu-miR6187. Additionally, the promoter of *HvGATA25* contains ABRE elements involved in abscisic acid response and MBS elements related to stress response regulation. Therefore, the *HvGATA25* gene may be integral to barley's adaptation to stressful environments. Additionaly, it has been reported that hvu-miR168-5p is down-regulated in barley under salt stress [7], and in this study, *HvGATA14* was regulated by hvu-miR168-5p, with its promoter containing DRE cis-elements responsive to stress conditions. hvu-miR5049e expression increased in barley under biotic stress [56], with *HvGATA8* under its regulation; hvu-miR5048a expression increased under salt stress [7], and hvu-miR5048b expression increased under biotic stress [57]. In the current study, *HvGATA16* was regulated by both of these miRNAs, and *HvGATA8* and *HvGATA16* both contained ABER and TC-rich repeats cis-elements. Among the 27 *HvGATAs*, only *HvGATA3* and *HvGATA5* lacked ABER cis-elements, while the remaining 25 *HvGATAs* were involved in abscisic acid responsiveness, highlighting the close association between *HvGATA* gene family and stress responses.

Conclusion

In summary, this research identified 27 members of the *HvGATA* gene family in the barley genome. The *HvGATA* genes in the barley genome exhibit a pattern similar to that of *GATA* genes in other plant species and are closely implicated in the regulatory mechanisms of abiotic stress responses. A thorough characterization of the barley *GATA* gene family was performed through an extensive analysis of its physicochemical properties, phylogenetic relationships, gene structure, chromosomal distribution, cis-regulatory elements, and expression patterns. These findings offer essential insights for the manipulation of *GATA* genes and the application of marker-assisted breeding in barley. However, further functional characterization of the genes is essential to reveal the precise functional attributes of *HvGATA* genes.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12864-025-11834-0.

Supplementary Material 1.

Authors' contributions

WL and WM conceived and designed the research. YZ completed the experiment and wrote the manuscript. RX and CY performed the experiments and data analysis. JZ, XD and DX revised the manuscript. All authors agreed to the published version of the manuscript.

Funding

This work was supported by the Joint Funds of the National Natural Science Foundation of China (No. U22A20457), and the High-Level Talents Project of Qingdao Agricultural University (663/1122023).

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 7 February 2025 / Accepted: 17 June 2025
Published online: 07 July 2025

References

1. Franco-Zorrilla J, López-Vidriero I, Carrasco J, Godoy M, Vera P, Solano R. DNA-binding specificities of plant transcription factors and their potential to define target genes. Proc Natl Acad Sci USA. 2014;111:2367–72.
2. Jin J, He K, Tang X, Li Z, Lv L, Zhao Y, Luo J, Gao G. An Arabidopsis transcriptional regulatory map reveals distinct functional and evolutionary features of novel transcription factors. Mol Biol Evol. 2015;32:1767–73.
3. Feng X, Yu Q, Zeng J, He X, Liu W. Genome-wide identification and characterization of GATA family genes in wheat. BMC Plant Biol. 2022;22:372.
4. Reyes J, Muro-Pastor M, Florencio F. The GATA family of transcription factors in Arabidopsis and rice. Plant Physiol. 2004;134:1718–32.
5. Chen H, Shao H, Li K, Zhang D, Fan S, Li Y, Han M. Genome-wide identification, evolution, and expression analysis of GATA transcription factors in Apple (*Malus×domestica* Borkh). Gene. 2017;627:460–72.
6. Zhang C, Hou Y, Hao Q, Chen H, Chen L, Yuan S, Shan Z, Zhang X, Yang Z, Qiu D, et al. Genome-wide survey of the soybean GATA transcription factor gene family and expression analysis under low nitrogen stress. PLoS ONE. 2015;10:e0125174.
7. Deng P, Wang L, Cui L, Feng K, Liu F, Du X, Tong W, Nie X, Ji W, Wei S. Global identification of MicroRNAs and their targets in barley under salinity stress. PLoS ONE. 2015;10(9):e0137990.
8. An Y, Zhou Y, Han X, Shen C, Wang S, Liu C, Yin W, Xia X. The GATA transcription factor GNC plays an important role in photosynthesis and growth in Poplar. J Exp Bot. 2020;71:1969–84.
9. Zhang Y, Zhang Y, Zhang L, Huang H, Yang B, Luan S, Xue H, Lin W. *OsGATA7* modulates brassinosteroids-mediated growth regulation and influences architecture and grain shape. Plant Biotechnol J. 2018;16(7):1261–4.
10. Zhu H, Zhai H, He S, Zhang H, Gao S, Liu Q. A novel Sweetpotato GATA transcription factor, IbGATA24, interacting with IbCOP9-5a positively regulates drought and salt tolerance. Environ Exp Bot. 2022;194:104735.
11. Liu H, Li T, Wang Y, Zheng J, Li H, Hao C, Zhang X. TaZIM-A1 negatively regulates flowering time in common wheat (*Triticum aestivum* L). J Integr Plant Biol. 2019;61:359–76.
12. Liu X, Zhu X, Wei X, Lu C, Shen F, Zhang X, Zhang Z. The wheat LLM-domain-containing transcription factor TaGATA1 positively modulates host immune response to *Rhizoctonia cerealis*. J Exp Bot. 2020;71:344–55.
13. Cai S, Shen Q, Huang Y, Han Z, Wu D, Chen Z, Nevo E, Zhang G. Multi-Omics analysis reveals the mechanism underlying the edaphic adaptation in wild barley at evolution slope (Tabigha). Adv Sci. 2021;8(20):e2101374.
14. Karunarathne S, Walker E, Sharma D, Li C, Han Y. Genetic resources and precise gene editing for targeted improvement of barley abiotic stress tolerance. J Zhejiang Univ Sci B. 2023;24(12):1069–92.
15. Taketa S, Amano S, Tsujino Y, Sato T, Saisho D, Kakeda K, Nomura M, Suzuki T, Matsumoto T, Sato K, et al. Barley grain with adhering hulls is controlled by an ERF family transcription factor gene regulating a lipid biosynthesis pathway. Proc Natl Acad Sci USA. 2008;105:4062–7.
16. Kersey P, Allen J, Allot A, Barba M, Boddu S, Bolt B, Carvalho-Silva D, Christensen M, Davis P, Grabmueller C, et al. Ensembl genomes 2018: an integrated omics infrastructure for non-vertebrate species. Nucleic Acids Res. 2018;46:D802–8.
17. Finn R, Clements J, Eddy S. HMMER web server: interactive sequence similarity searching. Nucleic Acids Res. 2011;39:W29–37.
18. Finn R, Coggill P, Eberhardt R, Eddy S, Mistry J, Mitchell A, Potter S, Punta M, Qureshi M, Sangrador-Vegas A, et al. The Pfam protein families database: towards a more sustainable future. Nucleic Acids Res. 2016;44:D279–285.
19. Artimo P, Jonnalagedda M, Arnold K, Baratin D, Csardi G, de Castro E, Duvaud S, Flegel V, Fortier A, Gasteiger E, et al. ExPASy: SIB bioinformatics resource portal. Nucleic Acids Res. 2012;40:W597–603.
20. Larkin M, Blackshields G, Brown NP, Chenna R, McGettigan P, McWilliam H, Valentin F, Wallace IM, Wilm A, Lopez R, et al. Clustal W and clustal X version 2.0. Bioinformatics. 2007;23:2947–8.
21. Kumar S, Stecher G, Tamura K. MEGA7: molecular evolutionary genetics analysis version 7.0 for bigger datasets. Mol Biol Evol. 2016;33:1870–4.
22. Letunic I, Bork P. Interactive tree of life (iTOL) v4: recent updates and new developments. Nucleic Acids Res. 2019;47:W256–9.
23. Voorrips R. MapChart: software for the graphical presentation of linkage maps and QTLs. J Hered. 2002;93:77–8.
24. Wang Y, Tang H, Debarry J, Tan X, Li J, Wang X, Lee T, Jin H, Marler B, Guo H, et al. MCScanX: a toolkit for detection and evolutionary analysis of gene synteny and collinearity. Nucleic Acids Res. 2012;40:e49.
25. Zhang Z, Xiao J, Wu J, Zhang H, Liu G, Wang X, Dai L. ParaAT: a parallel tool for constructing multiple protein-coding DNA alignments. Biochem Biophys Res Commun. 2012;419:779–81.
26. Wang D, Zhang Y, Zhang Z, Zhu J, Yu J. KaKs_Calculator 2.0: a toolkit incorporating gamma-series methods and sliding window strategies. Genom Proteom Bioinform. 2010;8:77–80.
27. Hu B, Jin J, Guo A, Zhang H, Luo J, Gao G. GSDS 2.0: an upgraded gene feature visualization server. Bioinformatics. 2015;31:1296–7.
28. Bailey T, Boden M, Buske F, Frith M, Grant Ce, Clementi L, Ren J, Li W, Noble W. MEME SUITE: tools for motif discovery and searching. Nucleic Acids Res. 2009;37:W202–208.
29. Chen C, Chen H, Zhang Y, Thomas H, Frank M, He Y, Xia R. TBtools: an integrative toolkit developed for interactive analyses of big biological data. Mol Plant. 2020;13:1194–202.
30. Lescot M, Déhais P, Thijs G, Marchal K, Moreau Y, Van de Peer Y, Rouzé P, Rombauts S. PlantCARE, a database of plant cis-acting regulatory elements and a portal to tools for in Silico analysis of promoter sequences. Nucleic Acids Res. 2002;30:325–7.
31. Tian F, Yang D, Meng Y, Jin J, Gao G. PlantRegMap: charting functional regulatory maps in plants. Nucleic Acids Res. 2020;48:D1104–13.
32. Su W, Raza A, Zeng L, Gao A, Lv Y, Ding X, Cheng Y, Zou X. Genome-wide analysis and expression patterns of lipid phospholipid phospholipase gene family in *Brassica napus* L. BMC Genomics. 2021;22:548.
33. Luo X, Lin W, Zhu S, Zhu J, Sun Y, Fan X, Cheng M, Hao Y, Oh E, Tian M, et al. Integration of light- and brassinosteroid-signaling pathways by a GATA transcription factor in *Arabidopsis*. Dev Cell. 2010;19:872–83.
34. Zhang H, Wang H, Zhu Q, Gao Y, Wang H, Zhao L, Wang Y, Xi F, Wang W, Yang Y, et al. Transcriptome characterization of Moso bamboo (*Phyllostachys edulis*) seedlings in response to exogenous Gibberellin applications. BMC Plant Biol. 2018;18:125.
35. Livak K, Schmittgen T. Analysis of relative gene expression data using real-time quantitative PCR and the 2(-Delta Delta C(T)) method. Methods. 2001;25:402–8.
36. Wang D, Zhang S, He F, Zhu J, Hu S, Yu J. How do variable substitution rates influence Ka and Ks calculations? Genomics Proteom Bioinf. 2009;7:116–27.
37. Swigonová Z, Lai J, Ma J, Ramakrishna W, Llaca V, Bennetzen J, Messing J. Close split of sorghum and maize genome progenitors. Genome Res. 2004;14(10a):1916–23.
38. Fujita Y, Fujita M, Satoh R, Maruyama K, Parvez M, Seki M, Hiratsu K, Ohme-Takagi M, Shinozaki K, Yamaguchi-Shinozaki K. AREB1 is a transcription activator of novel ABRE-dependent ABA signaling that enhances drought stress tolerance in *Arabidopsis*. Plant Cell. 2005;17(12):3470–88.
39. Wu G, Tian N, She F, Cao A, Wu W, Zheng S, Yang N. Characteristics analysis of early responsive to dehydration genes in Arabidopsis thaliana (*AtERD*). Plant Signal Behav. 2023;18(1):2105021.
40. Shen M, Zhao K, Luo X, Guo L, Ma Z, Wen L, Lin S, Lin Y, Sun H, Ahmad S. Genome mining of WOX-ARF gene linkage in *Machilus Pauhoi* underpinned cambial activity associated with IAA induction. Front Plant Sci. 2024;15:1364086.
41. Du X, Lu Y, Sun H, Duan W, Hu Y, Yan Y. Genome-wide analysis of wheat GATA transcription factor genes reveals their molecular evolutionary characteristics and involvement in salt and drought tolerance. Int J Mol Sci. 2022;24;2023.
42. Zinati Z, Nazari L. Deciphering the molecular basis of abiotic stress response in cucumber (*Cucumis sativus* L.) using RNA-Seq meta-analysis, systems biology, and machine learning approaches. Sci Rep. 2023;13(1):12942.
43. Cao Y, Tanaka K, Nguyen C, Stacey G. Extracellular ATP is a central signaling molecule in plant stress responses. Curr Opin Plant Biol. 2014;20:82–7.

44. Yetgin A. Exploring the dynamic nature of root plasticity and morphology in the face of changing environments. Front Ecol Environ. 2024;44:112–9.
45. Gao Y, Su Y, Chao D. Exploring the function of plant root diffusion barriers in sealing and shielding for environmental adaptation. Nat Plants. 2024;10:1865–74.
46. Liu P, Koizuka N, Martin R, Nonogaki H. The *BME3 (Blue micropylar end 3)* GATA zinc finger transcription factor is a positive regulator of Arabidopsis seed germination. Plant J. 2005;44(6):960–71.
47. Ravindran P, Verma V, Stamm P, Kumar P. A novel RGL2-DOF6 complex contributes to primary seed dormancy in *Arabidopsis thaliana* by regulating a GATA transcription factor. Mol Plant. 2017;10(10):1307–20.
48. Richter R, Bastakis E, Schwechheimer C. Cross-repressive interactions between SOC1 and the GATAs GNC and GNL/CGA1 in the control of greening, cold tolerance, and flowering time in Arabidopsis. Plant Physiol. 2013;162(4):1992–2004.
49. Zhang X, Ma J, Yang S, Yao W, Zhang N, Hao X, Xu W. Analysis of GATA transcription factors and their expression patterns under abiotic stress in grapevine (*Vitis vinifera* L). BMC Plant Biol. 2023;23(1):611.
50. Zhu W, Guo Y, Chen Y, Wu D, Jiang L. Genome-wide identification, phylogenetic and expression pattern analysis of *GATA* family genes in *Brassica napus*. BMC Plant Biol. 2020;20(1):543.
51. Nutan K, Singla-Pareek S, Pareek A. The *Saltol* QTL-localized transcription factor OsGATA8 plays an important role in stress tolerance and seed development in Arabidopsis and rice. J Exp Bot. 2020;71(2):684–98.
52. Zhang H, Wu T, Li Z, Huang K, Kim N, Ma Z, Kwon S, Jiang W, Du X. OsGATA16, a GATA transcription factor, confers cold tolerance by repressing *OsWRKY45-1* at the seedling stage in rice. Rice. 2021;14(1):42.
53. Wang Y, Cao X, Zhang D, Li Y, Wang Q, Ma F, Xu X, Zhan X, Hu T. SlGATA17, A tomato GATA protein, interacts with SlHY5 to modulate salinity tolerance and germination. Environ Exp Bot. 2023;206:105191.
54. Bartel D. MicroRNAs: target recognition and regulatory functions. Cell. 2009;136(2):215–33.
55. Mutum R, Kumar S, Balyan S, Kansal S, Mathur S, Raghuvanshi S. Identification of novel MiRNAs from drought tolerant rice variety nagina 22. Sci Rep. 2016;6:30786.
56. Yao X, Wang Y, Yao Y, Bai Y, Wu K, Qiao Y. Identification MicroRNAs and target genes in Tibetan hulless barley to BLS infection. Agron J. 2021;113:2273–92.
57. Habachi-Houimli Y, Khalfallah Y, Mezghani-Khemakhem M, Makni H, Makni M, Bouktila D. Genome-wide identification, characterization, and evolutionary analysis of NBS-encoding resistance genes in barley. 3 Biotech. 2018;8:453.

Publisher’s Note